Chapter 4

4. Factor Analysis.

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Introduction

Consider a random vector $X = (X_1, \ldots, X_p)'$.

The goal is to find a simple model which describes the dependence between the components.

The target is similar as in PCA, namely to find a lower dimensional structure which can help interpret the data.

Introduction

A factor model assumes that components of X are driven by a set of latent factors $F = (F_1, \ldots, F_k)'$.

Here k, the number of factors, is typically much smaller than p.

The relation is linear, so that

$$X \approx \Lambda F \quad \text{where } \Lambda \in \mathbb{R}^{p \times k}.$$

Besides these common factors it is assumed that there is some idiosyncratic noise attached to each components, so that we get the relation

$$X = \Lambda F + U, \quad \text{where } \mathrm{Var}(U) = \mathrm{diag}.$$

Introduction

So the i-th component of X is given by

$$X_i = \Lambda_{i1} F_1 + \cdots + \Lambda_{ik} F_k + U_i, \quad 1 \leq i \leq p.$$

The blue part is the common component.

The red part is the idiosyncratic component.

Famous examples of factor models

The famous CAPM (Capital Asset Pricing Model) is a single factor model. It models the returns of n risky assets. The latent factor is the market rate of return.

The OCEAN Model is used in psychology. According to that model five latent personality traits (Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism) explain the personality of a person.



Model and properties

Definition

Let X be a p-r.v. with $EX = \mu$ und $\mathrm{Var}(X) = \Sigma$. We say that the k-factor model *holds for X, if*

$$X - \mu = \Lambda F + U,$$

where Λ is a $p \times k$-matrix, $F = (F_1, \ldots, F_k)'$ and $U = (U_1, \ldots, U_p)'$ are random vectors with

$$\begin{aligned} E(F) &= 0, \quad \mathrm{Var}(F) = I_k \\ E(U) &= 0, \quad \mathrm{Var}(U) = \Psi = \mathrm{diag}(\psi_{11}, \ldots, \psi_{pp}) \end{aligned}$$

and $E(FU') = 0$.

Model and properties

We conclude that

$$\Sigma = \Lambda\Lambda' + \Psi,$$

and thus

$$\mathrm{Var}(X_i) = \sum_{j=1}^{k} \Lambda_{ij}^2 + \psi_{ii} = h_i^2 + \psi_{ii}.$$

The matrix Λ is called loading matrix.

The part of the variance that is due to the common factors – h_i^2 – is called communality of X_i.

In contrast, ψ_{ii}, which is the part of the variance that is specific to the i-th component, is called the individual/specific variance of X_i.

Note also that $\mathrm{Cov}(X, F) = \Lambda$.

The components of F are called factor scores.

Model and properties

One can show that the k-factor model is in fact solely a model for the covariance structure of X.

Proposition

The k-factor model for X holds, iff

$$\mathrm{Var}(X) = \Lambda\Lambda' + \Psi.$$

Model and properties

Let $C = \mathrm{diag}(c_1, \ldots, c_p)$ be a scaling matrix and d a p-vector. If the k-factor model holds for X, then it also holds for $Y = CX + d$:

$$\begin{aligned} \mathrm{Var}(Y) &= C\mathrm{Var}(X)C \\ &= C\Lambda_X\Lambda_X'C + C\Psi_X C \\ &= \Lambda_Y\Lambda_Y' + \Psi_Y, \end{aligned}$$

with $\Lambda_Y = C\Lambda_X$ and $\Psi_Y = C\Psi_X C$. $(C = C')$.

This implies, for example, that we can also base the factor model on the correlation matrix.

Model and properties

The loading matrix is not unique. Let G be an orthogonal $k \times k$-matrix. Since $GG' = I_k$ it follows that

$$X - \mu = (\Lambda G)(G'F) + U.$$

Furthermore $E(G'F) = 0$ und $\mathrm{Var}(G'F) = I_k$.

This is in general circumvent by imposing some side condition: e.g.

$$(*) \qquad \Lambda'\Psi^{-1}\Lambda \quad \text{is diagonal.}$$

We will use this freedom to rotate, to achieve easier interpretation.

Estimation

We wish to estimate Λ and Ψ.

The general strategy is to replace Σ by $\hat{\Sigma}_n$ and to find $\hat{\Lambda}$ and $\hat{\Psi}$ such that (at least approximatively)

$$\hat{\Sigma}_n = \hat{\Lambda}\hat{\Lambda}' + \hat{\Psi}.$$

The two main methods for estimation are principal factor analysis and maximum likelihood factor analysis.

In general it is recommended to use ML. For high dimensional data, also PCA approach is suitable as well.

Estimation

Note that the model has $p(k+1)$ parameters, the constraint we impose leaves $p(k+1) - k(k-1)/2$ free parameters.

Hence, a dimension reduction is taking place only if

$$s = p(p+1)/2 - \{p(k+1) - k(k-1)/2\} > 0.$$

Estimation of factor scores

We recall that the factor scores F are not observed, and the goal is now to get a proxy from the observed data.

Once Λ is known or estimated, the factor model resembles formally a linear model $X = \Lambda F + U$ (observation-wise).

We can now obtain an estimate for the factor scores by least-squares, or better by weighted least squares:

$$\underbrace{\Psi^{-1/2}X}_{Y} = \underbrace{\Psi^{-1/2}\Lambda}_{W} F + \underbrace{\Psi^{-1/2}U}_{\epsilon \text{ with } \mathrm{Var}(\epsilon) = \mathrm{diag}(p)},$$

yielding Bartlett's factor scores

$$\hat{F} = (W'W)^{-1}W'Y = (\Lambda'\Psi^{-1}\Lambda)^{-1}\Lambda'\Psi^{-1}X.$$

Return data

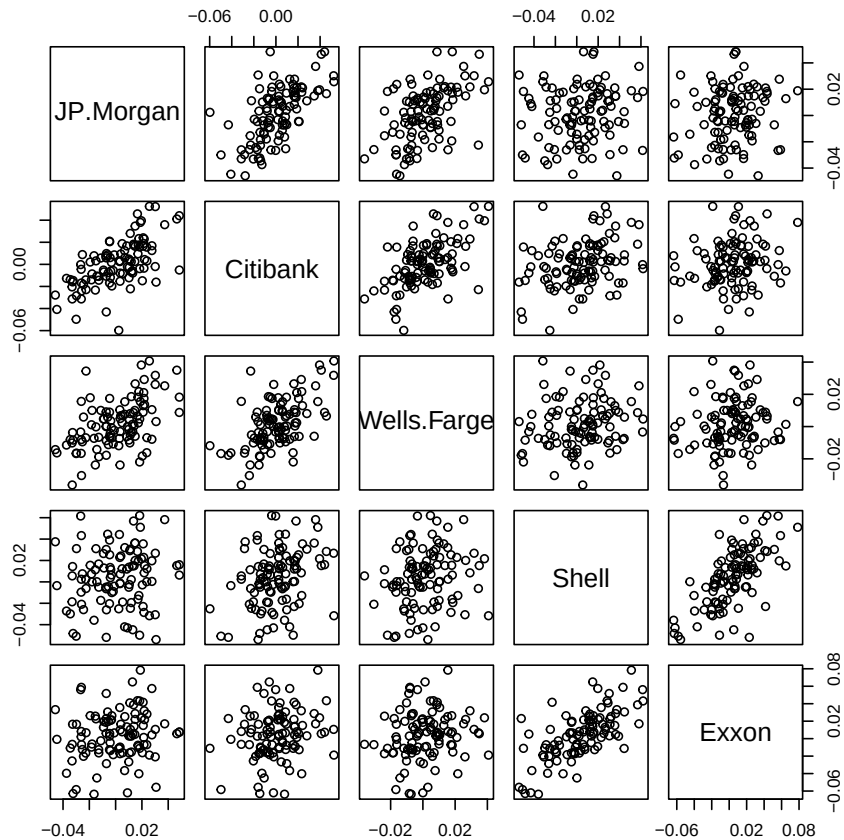
In the following example we analyse weekly returns for 5 stocks listed on NYSE:

3 from the bank sector (JP Morgan, Citibank, Wells Fargo) and two from energy sector (Shell, Exxon).

```
> s.names <- c("JP Morgan","Citibank","Wells Farge","Shell","Exxon")
> stock <- read.table("Data/T8-4.dat",col.name=s.names)
> head(stock)
   JP.Morgan   Citibank Wells.Farge      Shell      Exxon
1  0.0130338 -0.0078431  -0.0031889 -0.0447693  0.0052151
2  0.0084862  0.0166886  -0.0062100  0.0119560  0.0134890
3 -0.0179153 -0.0086393   0.0100360  0.0000000 -0.0061428
4  0.0215589 -0.0034858   0.0174353 -0.0285917 -0.0069534
5  0.0108225  0.0037167  -0.0101345  0.0291900  0.0409751
6  0.0101713 -0.0121978  -0.0083768  0.0137083  0.0029895
 ....
```

Return data

Visual inspection of the scatterplot indicates that we have comparably strong correlation within sectors, but small correlation across sectors.



Return data

This is confirmed by a look at the correlation matrix.

```
> options(digits=2)
> cor(stock)
             JP.Morgan Citibank Wells.Farge Shell Exxon
JP.Morgan         1.00     0.63         0.51  0.11  0.15
Citibank          0.63     1.00         0.57  0.32  0.21
Wells.Farge       0.51     0.57         1.00  0.18  0.15
Shell             0.11     0.32         0.18  1.00  0.68
Exxon             0.15     0.21         0.15  0.68  1.00
```

It seems natural to try a 2-factor model. The guess is that we can interpret the two factors as *bank factor* and *energy factor*.

Return data

```
> fa <- factanal(stock,factors=2)
> print(fa)

Call:
factanal(x = stock, factors = 2)

Uniquenesses:
  JP.Morgan    Citibank Wells.Farge       Shell       Exxon
      0.417       0.275       0.542       0.005       0.530

Loadings:
            Factor1 Factor2
JP.Morgan   0.763
Citibank    0.819   0.232
Wells.Farge 0.668   0.108
Shell       0.113   0.991
Exxon       0.108   0.677
```

Return data

```
                Factor1 Factor2
SS loadings       1.73    1.51
Proportion Var    0.34    0.30
Cumulative Var    0.34    0.65

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 2 on 1 degree of freedom.
The p-value is 0.16
```

From the loading matrix we see that large values and small values of the factor loadings are complementary for bank related stocks and energy stocks. We can interpret the factors accordingly.

Return data

By default `factanal` uses

1. ML factor analysis
2. it works with the correlation matrix
3. applies a *varimax* rotation

Note that $|\hat{\Psi}| \approx 0$, which is the so-called *Heywood case*. Could be a sign that too many factors were used.

Return data

Explanation of the output:

- `uniqueness` corresponds to $(\hat{\psi}_1, \ldots, \hat{\psi}_5) = (1 - \hat{h}_1^2, \ldots, 1 - \hat{h}_5^2)$.
- The columns `Factor1 Factor2` define $\hat{\Lambda}$.
- `SS loadings` corresponds to $\sum_{i=1}^5 \hat{\lambda}_{i1}^2$ and $\sum_{i=1}^5 \hat{\lambda}_{i2}^2$.
- `Proportion Var` is equal to $\hat{\sum}_{i=1}^5 \hat{\lambda}_{i1}^2/5$ and $\sum_{i=1}^5 \hat{\lambda}_{i2}^2/5$ (since correlation is used). In general it is $\sum_{i=1}^5 \hat{\lambda}_{ij}^2/tr(\hat{\Sigma}_n)$.

Return data

The implemented test shows that we can accept H_0: *the number of factors equals 2*.

Below are the residuals $R - (\hat{\Lambda}\hat{\Lambda}' + \hat{\Psi})$.

	JP.Morgan	Citibank	Wells F.	Shell	Exxon
JP.Morgan	-1.05e-07	-7.49e-06	2.56e-03	3.32e-04	-5.19e-02
Citibank	-7.49e-06	-3.25e-08	-1.60e-03	-2.11e-04	3.30e-02
Wells F.	2.56e-03	-1.60e-03	-5.15e-08	9.51e-06	-5.54e-04
Shell	3.32e-04	-2.11e-04	9.51e-06	1.55e-06	-1.21e-04
Exxon	-5.19e-02	3.30e-02	-5.54e-04	-1.21e-04	-2.67e-07

Rotation of the loadings

As it was already pointed out, the factor model is only unique up to rotations, and we are free to choose rotations that give the easiest interpretation (usually loadings should be close to 0 or 1).

Rotation of the loadings

Example: Lawley and Maxwell present the sample correlation matrix of examination scores in $p = 6$ subject areas (Gaelic, English, History, Arithmetic, Algebra, Geometry) for $n = 220$ male students.

$$R = \begin{pmatrix} 1 & 0.439 & 0.410 & 0.288 & 0.329 & 0.248 \\ & 1 & 0.351 & 0.354 & 0.320 & 0.329 \\ & & 1 & 0.164 & 0.190 & 0.181 \\ & & & 1 & 0.595 & 0.470 \\ & & & & 1 & 0.464 \\ & & & & & 1 \end{pmatrix}.$$

Rotation of the loadings: none

```
> factanal(covmat=R,factors=2,rotation="none")

Call:
factanal(factors = 2, covmat = R, rotation = "none")

Uniquenesses:
[1] 0.51 0.59 0.64 0.38 0.43 0.63

Loadings:
     Factor1 Factor2
[1,]  0.55    0.43
[2,]  0.57    0.29
[3,]  0.39    0.45
[4,]  0.74   -0.27
[5,]  0.72   -0.21
[6,]  0.59   -0.13

               Factor1 Factor2
SS loadings       2.21    0.61
Proportion Var    0.37    0.10
Cumulative Var    0.37    0.47

The degrees of freedom for the model is 4 and the fit was 0.0109
```

Rotation of the loadings: varimax

```
> factanal(covmat=R,factors=2)

Call:
factanal(factors = 2, covmat = R)

Uniquenesses:
[1] 0.51 0.59 0.64 0.38 0.43 0.63

Loadings:
     Factor1 Factor2
[1,] 0.235   0.659
[2,] 0.323   0.549
[3,]         0.590
[4,] 0.771   0.170
[5,] 0.724   0.213
[6,] 0.572   0.210

               Factor1 Factor2
SS loadings       1.61    1.20
Proportion Var    0.27    0.20
Cumulative Var    0.27    0.47

The degrees of freedom for the model is 4 and the fit was 0.0109
```

Rotation of the loadings: 20 degrees

```
> fa <- factanal(covmat=R,factors=2,rotation="none")
> load <- matrix(fa$loadings[1:12],ncol=2)
> rot <- function(load,angle){
+ O <- t(matrix(c(cos(angle),sin(angle),-sin(angle),cos(angle)),ncol=2))
+ load <- load%*%O
+ load
+ }
> rot(load,angle=pi/9)
     [,1]    [,2]
[1,] 0.37  0.5920
[2,] 0.44  0.4653
[3,] 0.21  0.5570
[4,] 0.79 -0.0031
[5,] 0.75  0.0490
[6,] 0.60  0.0799
```

Temperature data

In this next illustrative example we consider daily temperature data in Canadian weather stations. So we have a very high dimension ($p = 365$).

We consider two scenarios:

1. A spatial sample: temperature curves from different stations across the country ($n = 139$).
2. A temporal sample: temperature in Montreal across several years ($n = 33$).

Temperature data

Both times we fit a factor model.

The goal is to separate signal and noise (common component and idiosyncratic component) :

$$X = \underbrace{\Lambda F}_{\text{signal}} + \underbrace{U}_{\text{noise}}.$$

Temperature data

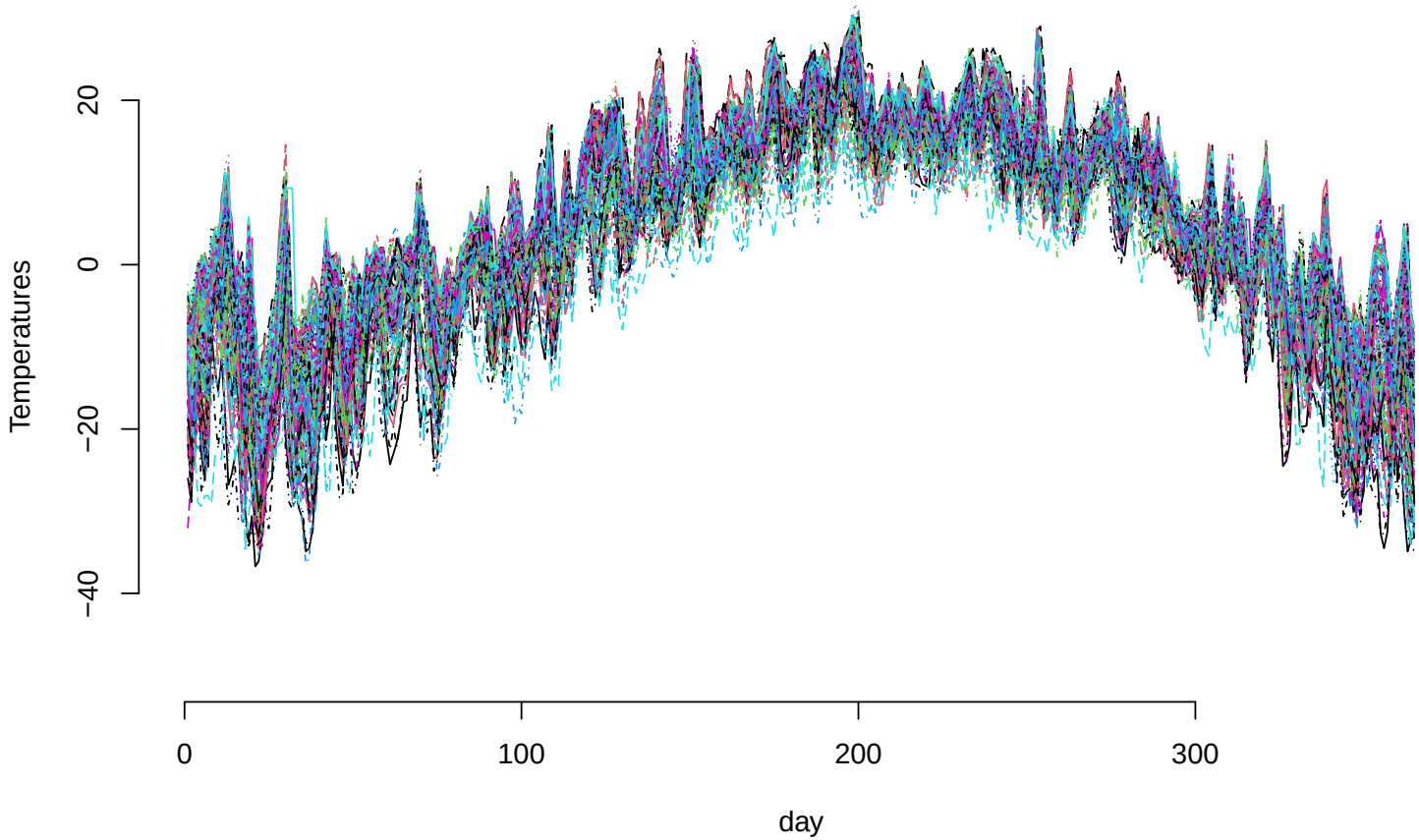


Figure 1: Canadian temperatures (spatial sample).

Temperature data

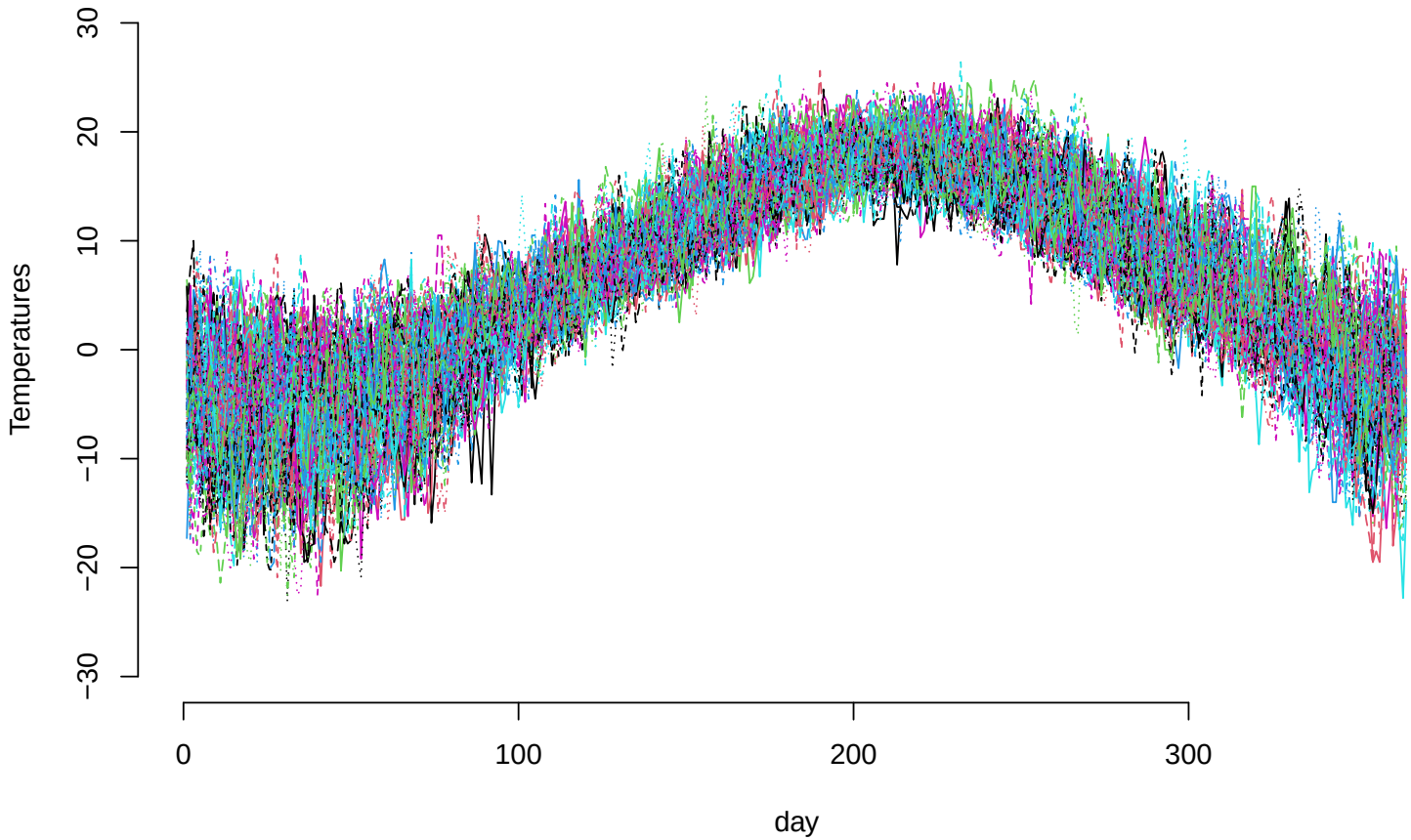


Figure 2: Montreal temperatures (temporal sample).

Temperature data

We fit a factor model with PCA method. With screeplots we determine number of factors

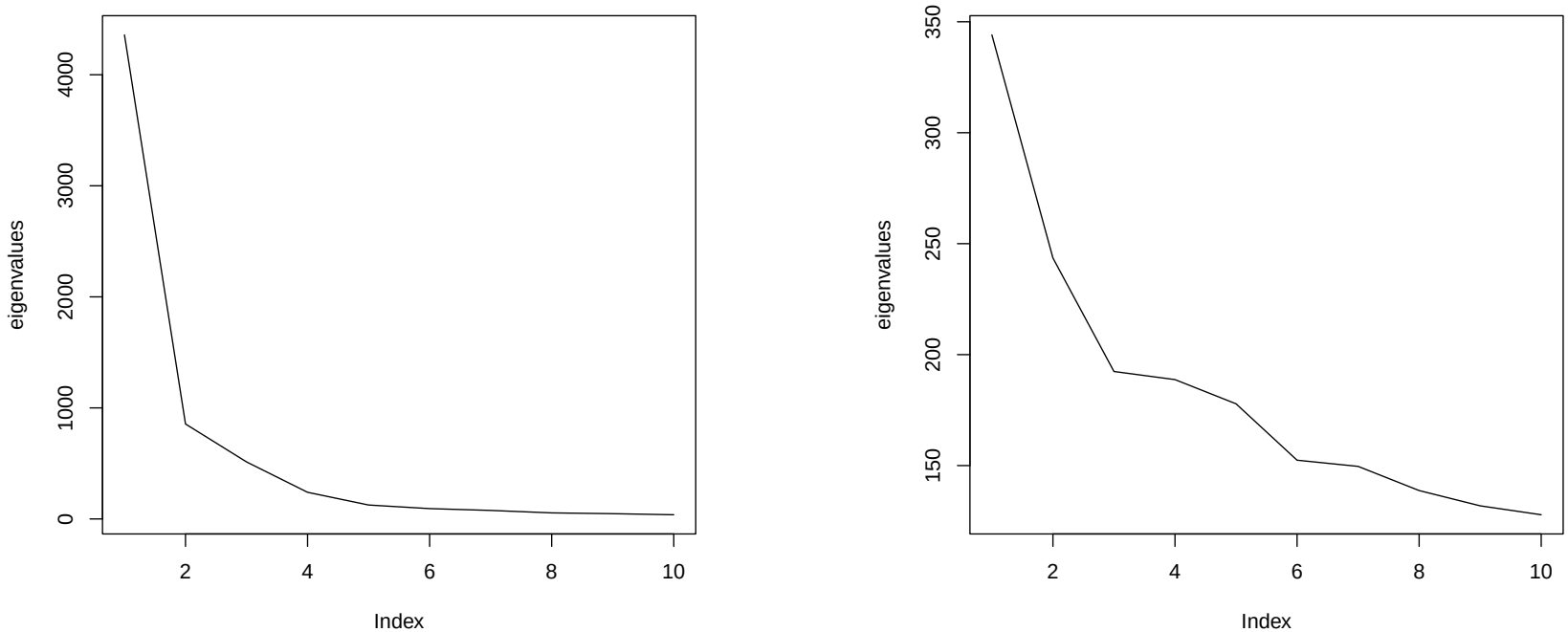


Figure 3: We choose 3 factors for the spatial data (left) and 4 for the temporal data (right).

Temperature data

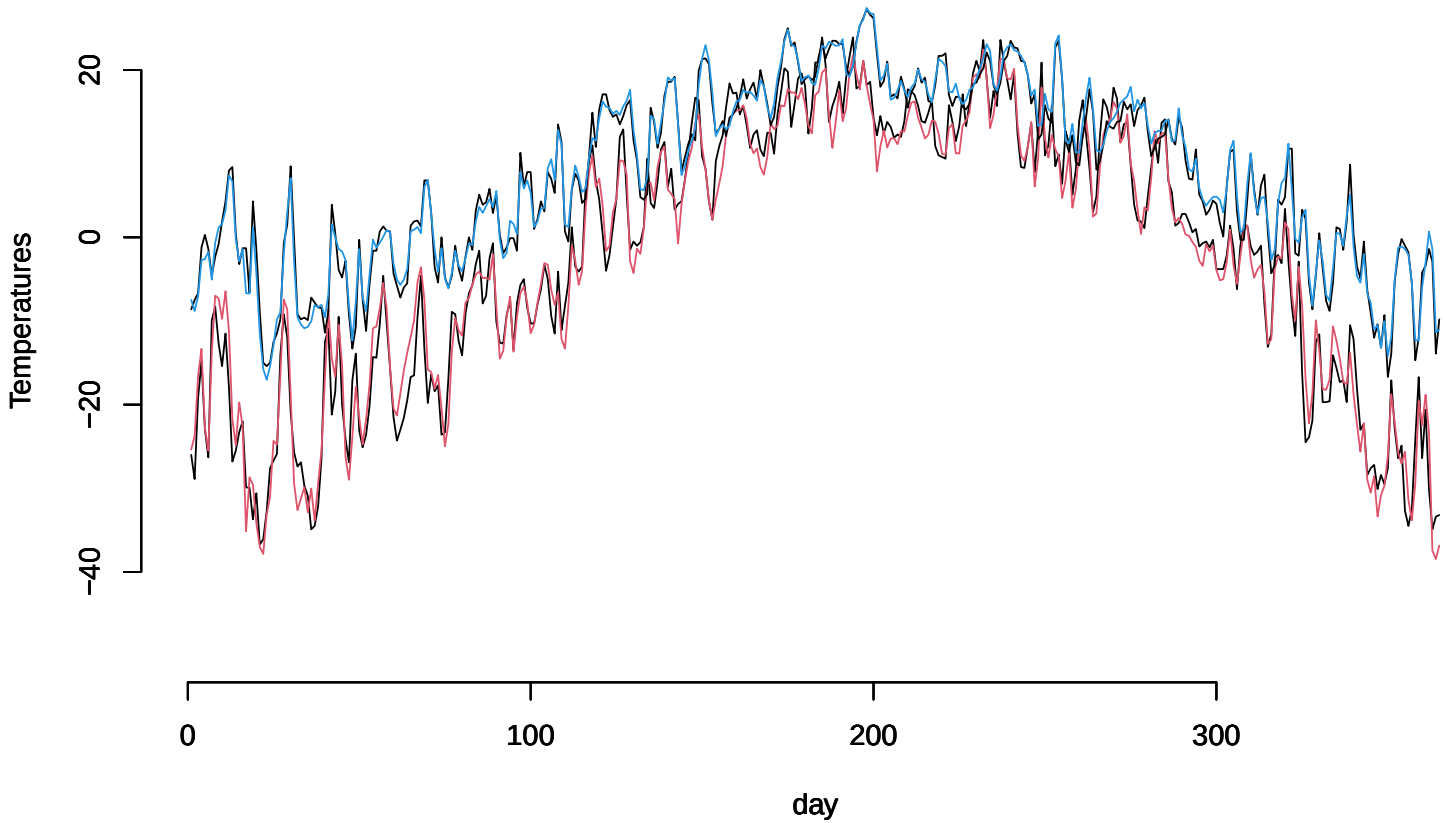


Figure 4: Spatial setup: We choose two stations (say index (i_1, i_2)) and estimate the "signals" ΛF_{i_1} and ΛF_{i_2} with 3 factors. Black lines are original data and red and blue the reconstructions.

Temperature data

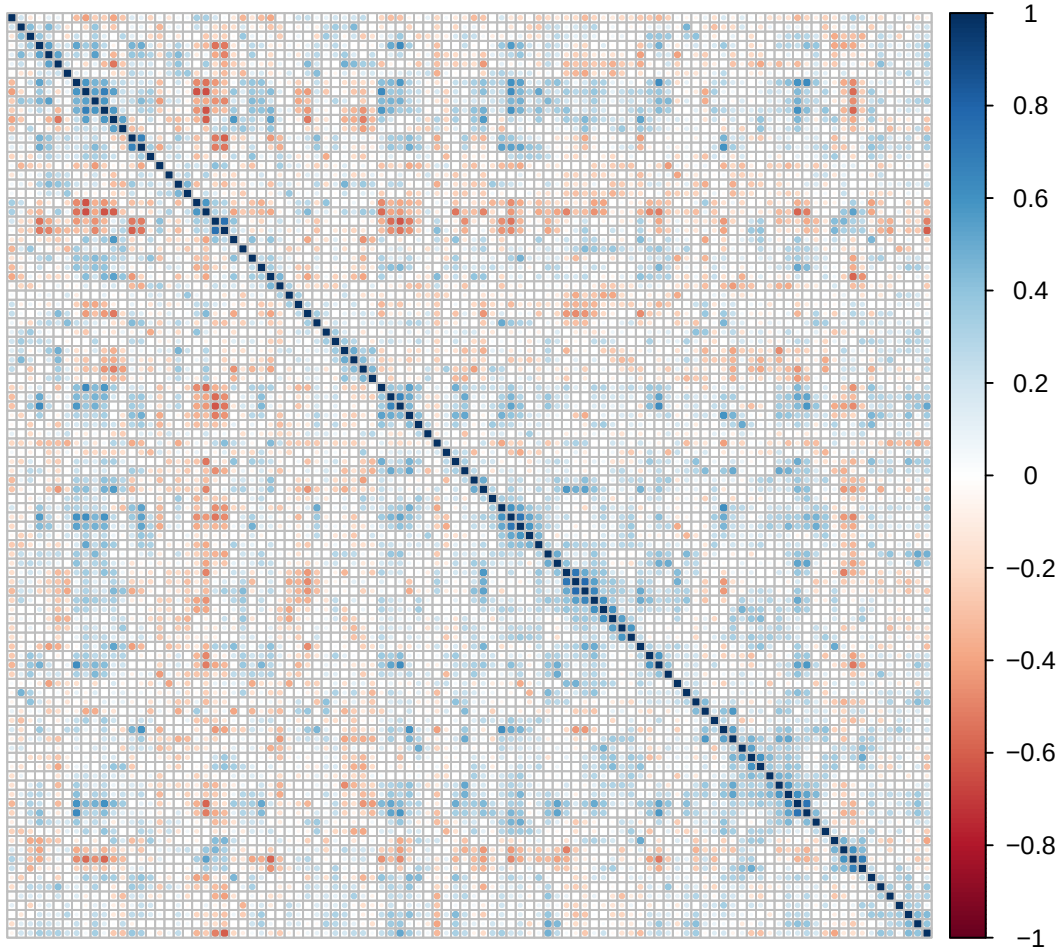


Figure 5: Spatial setup: Heat map of residual correlation matrix. For better visibility we choose only the first 100 days.

Temperature data

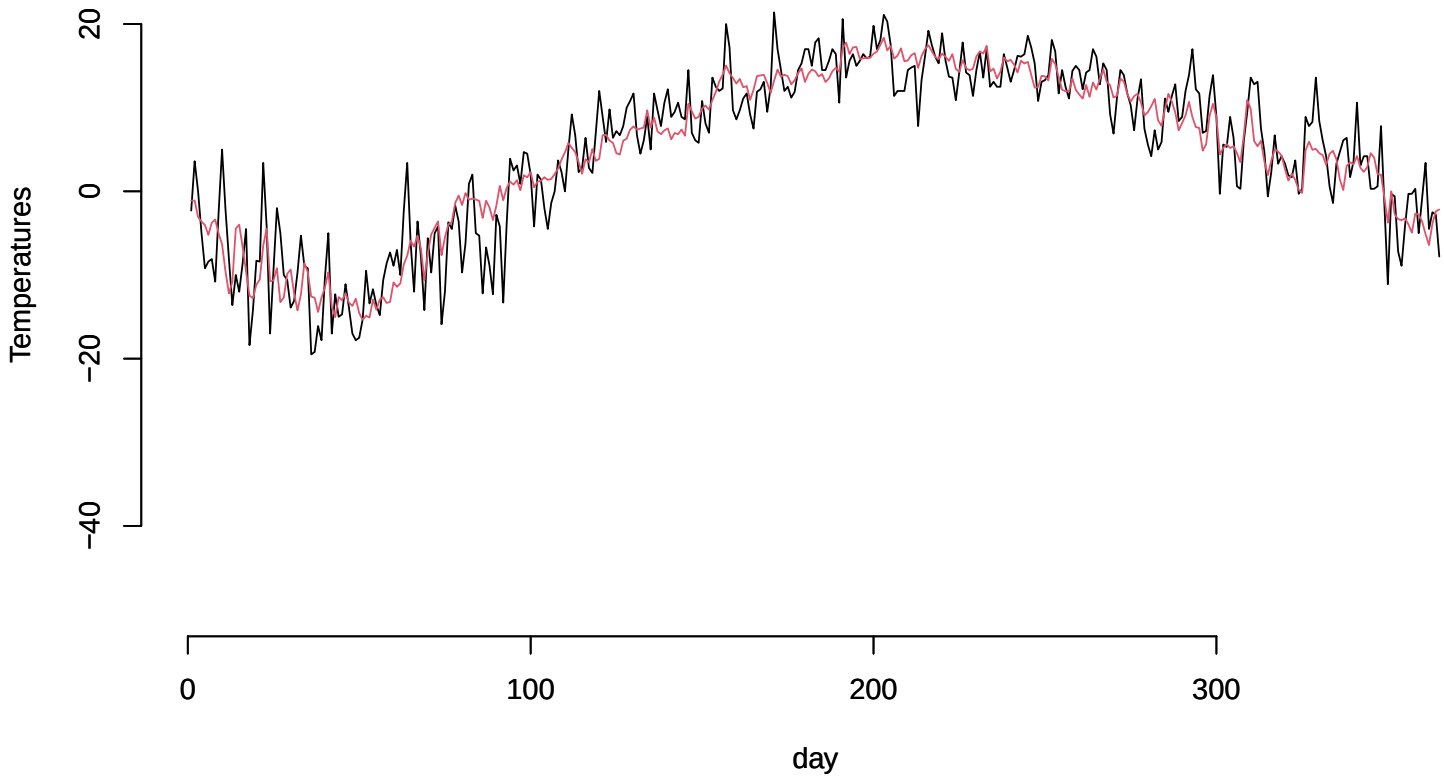


Figure 6: Temporal setup: We choose one year (say index t_1) and estimate the "signal" ΛF_{t_1} with 4 factors.

Temperature data

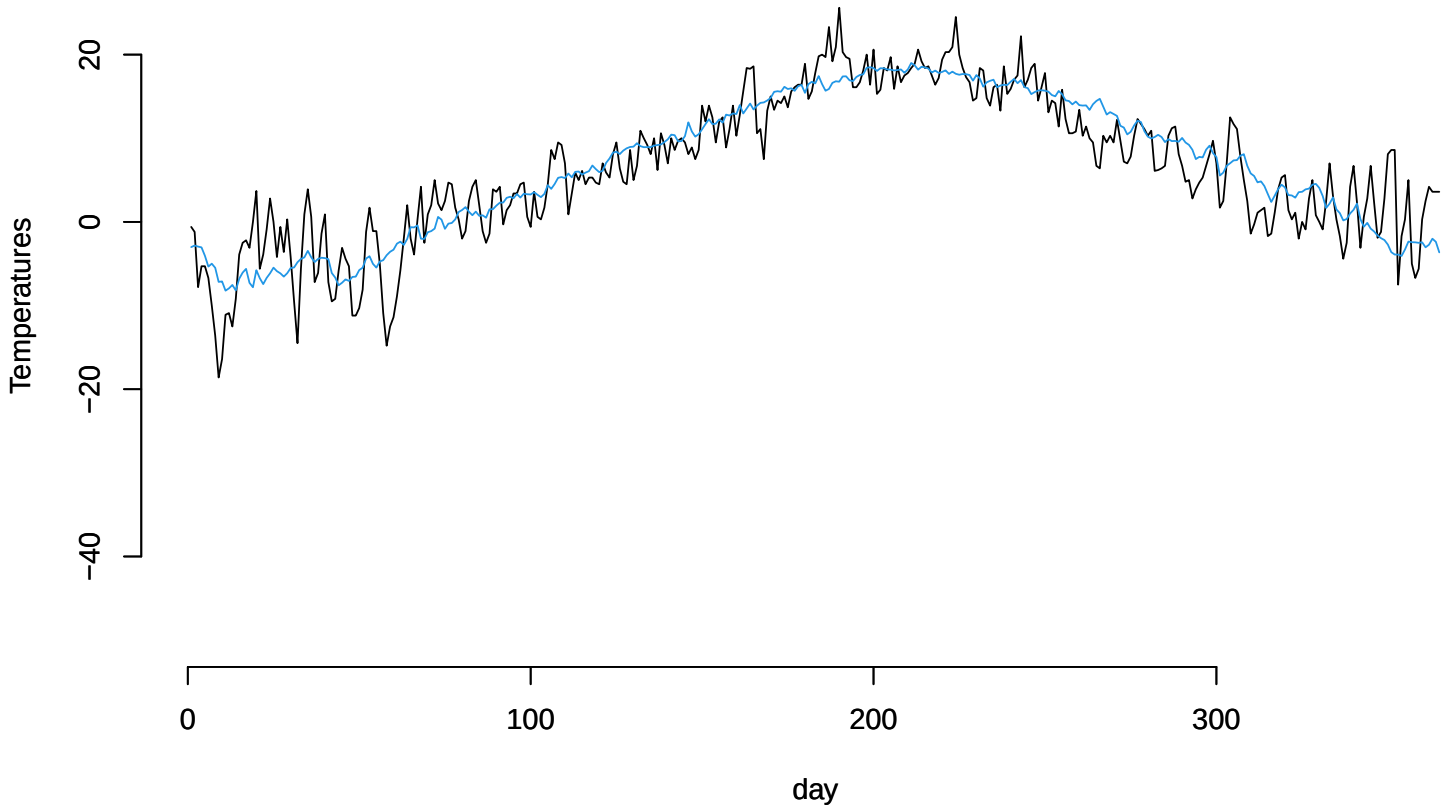


Figure 7: Analog to previous graph with another year t_2.

Temperature data

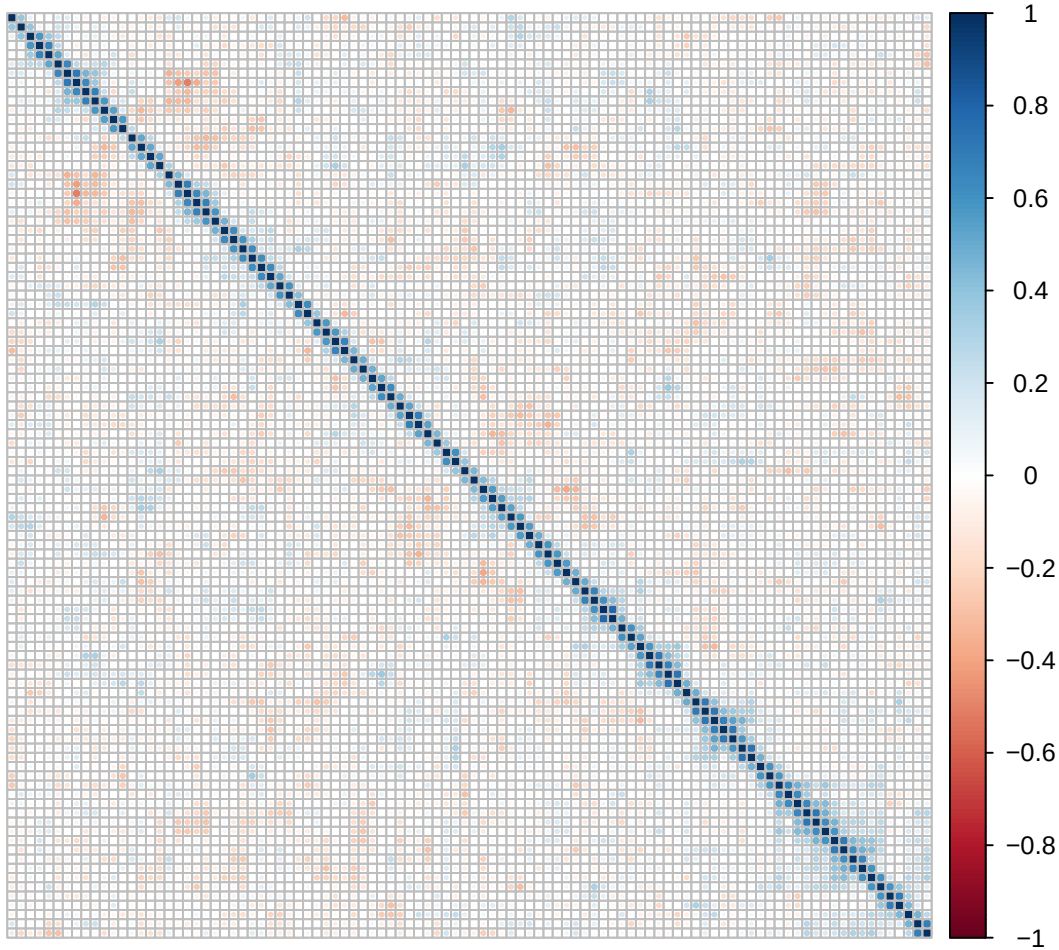


Figure 8: Temporal setup: Heat map of residual correlation matrix. (We choose only the first 100 days.)